



An anisotropic nonlinear multiscale damage model for mechanical behavior of a composite laminate with matrix cracks

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Abstract The first observed damage in a composite laminate is the matrix cracks. Although not leading to rupture, these cracks reduce the equivalent mechanical properties, leading to nonlinear stress–strain behavior. Because of the crack shielding effect (CSE), damage progression and stiffness reduction are nonlinear. Talreja has introduced an effective damage method for laminates with matrix cracks, and Singh has proposed a nonlinear model based on the method to assess stiffness reduction considering the CSE. But as Singh claimed, the model suffers from errors not addressed yet. This paper explains why Singh’s model has such limitations and presents a new anisotropic nonlinear multiscale damage model based on Talreja’s method. The model combines macro-scale continuum damage mechanics (CDM) with finite-element (FE) micromechanics. A glass/epoxy cross-ply laminate under tension is chosen as the case study, a light-transmission experiment is conducted to specify the damage progression, and an energy-based model is used to characterize it. As Talreja’s method suggests, a new function is proposed for the Helmholtz free energy, and the equations of mechanical properties in terms of the crack density are derived. FE micromechanical models with periodic boundary conditions (PBC) are developed to determine

the damage constants. The model is then applied to predict the stress–strain diagram of the laminate, the results are validated by tensile and loading–unloading experiments, and compared with Singh and Talreja’s models. Also, reference state selection criteria are established for the new model to obtain the best predictions, which was impossible with Singh’s model.

Keywords Continuum damage mechanics · Micromechanics · Energy-based model · Crack shielding effect · Matrix crack

1 Introduction

A composite laminate composed of unidirectional layers subjected to tensile or fatigue load develops matrix cracks (transverse cracks) as the first damage mechanism occurs in transversely oriented layers to the loading direction. These cracks usually grow very unstable and fast throughout the thickness and width of the layer and stop at the interface of the layers containing cracks and those without them. Therefore, since they do not lead to rupture, it is possible to continue loading and generate more. The matrix cracks are far from each other at the beginning of loading, and the decrease in mechanical properties caused by them has a linear relation with their number. As the load increases, their number increases, and they get closer to each other. As a result, the perturbation of the stress field around a crack has an effect on the

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opening displacement of adjacent cracks (COD) so the mechanical properties of the composite change nonlinearly with the number of cracks. This effect is called the crack shielding effect (CSE). As stated, matrix cracks do not lead to rupture but reduce the equivalent mechanical properties of the entire composite, which can cause loss of functionality in some structures. The second type of damage is secondary cracks. These small cracks start from the ends of the transverse cracks and grow along the interface of the layers when the laminate is under higher loads. By increasing the load, some of these cracks join together, leading to local delamination. This type of delamination is highly local and unstable and leads to a complete rupture of the structure. The latter two usually occur almost next to laminate rupture, mostly avoided in design, so the paper focus on the matrix cracks as the damage mechanism.

There are many methods to study damage mechanics. But the most-used ones are continuum damage mechanics (CDM) [1, 2] and micromechanics [3–6]. However, they still have limitations; In the case of CDM, each equation has a number of material constants related to damage. These constants should be determined through experimental tests. The more precise and complex the model becomes, the more constants appear in the relations. As a result, a very large number of experiments are needed to determine these constants. Finite element (FE) micromechanical models do not have this problem. But when a parameter changes in a numerical model, new modeling should be done. Therefore, in a study with several variables, such as geometry, layup, material type, loading, and damage, a large number of simulations must be performed. To overcome these limitations, Talreja introduced a new method [7]. In this modeling method, the relation between the mechanical properties of the laminate and matrix crack damage is determined based on CDM. Then, the material constants produced in the equations are obtained using the FE micromechanical models. With this method, there is no need to do many experiments or simulations. The other reliable ability of the model is that the constants obtained from FE micromechanical data for one laminate, as a reference laminate, can be used to predict stiffness changes for other laminates if the composite constituents are the same. This procedure is done by definition of the constraint parameters and relative constraint parameter. These parameters are computed by doing FE micromechanical simulations

and computing the averaged (over crack) CODs of the laminates.

Varna [8] calculated CDM material constants for an arbitrary laminate by a micromechanical model and checked out if Talreja's model is aligned with it. Varna found out that linear model of Talreja predicts larger stiffness degradation, and neglects crack sliding movement (CSM) which can be neglected only for the off-axis ply angle of close to 90. Since then, Talreja's method has been continuously developed and used in various studies; For example, fatigue study [9], viscoelastic response [10], multidirectional cracks [11], CSE consideration [12], progressive failure [13], multiaxial loading [14], low-velocity impact damage [15] and investigation of the performance of helicopter blades [16].

As stated, Talerja [7] presented a linear damage mechanics model based on the proposed method to predict the stiffness reduction of the symmetrical and balanced laminates with matrix cracks. However, this model did not have the ability to predict the nonlinear reduction of stiffness. Therefore, Singh [12] proposed a nonlinear model based on Talerja's method. Furthermore, Singh showed that although the nonlinear models inherit the abilities of Talreja's model, such as using reference laminate and good accuracy, it is accompanied by some errors. However, Singh did not address the problem properly. In this paper, these abandoned problems that cause errors in Singh's nonlinear model are addressed. Also, a new nonlinear analytical model is presented based on Talerja's method to diminish the mentioned errors, and the relations of mechanical properties are derived in terms of damage parameters. In doing so, a new function is proposed for the Helmholtz free energy of the composite, and the appropriate equations are derived. It is necessary to investigate the damage progression experimentally; There are numerous experimental methods to detect and measure matrix cracks, such as edge replication [17], light transmission [18], digital image capturing [19, 20], acoustic emission [21], X-ray radiography [22], scanning electron microscopy [23], and digital image correlation [24, 25]. For the case-study laminate, the most cost-efficient and fastest method is the light-transmission. So, this experimental test is conducted to investigate the damage progression in a glass/epoxy cross-ply laminate, and an energy-based model is used to characterize the progression. Finally, FE micromechanical models with periodic boundary conditions (PBC) are developed to determine the damage constants. The model has then validated against the quasi-static loading–unloading (LU)

experimental tests as well as Talerja and Singh's models. Moreover, the proposed model is applied to predict the nonlinear stress–strain diagram of the laminate under tension. Then the result is further validated by tensile experiments.

2 Experimental tests

To validate the proposed model, a glass/epoxy (GE) $[0/90_4]_s$ laminate is used in this paper. So, the mechanical properties of unidirectional GE layers, the diagram of elasticity modulus E_x in terms of applied strain ε , the diagram of applied strain ε in terms of crack density $1/s$ (number of cracks per unit length), and the stress–strain $\sigma_x - \varepsilon_x$ diagram should be determined through the experimental tests.

Tests to determine mechanical properties are conducted according to the ASTM D3039 and ASTM D5379 standards, and the results are presented in Table 1. To experimentally determine the $E_x - \varepsilon$ diagram, the quasi-static uniaxial loading–unloading (LU) tensile test in the direction of x is performed on the samples with a length of 250mm, width of 25mm, and thickness of 3mm (~ 0.3 mm for each layer). In this test, the sample is subjected to uniaxial quasi-static tension until rupture;

However, for some values of ε , the loading is interrupted, and a percentage of it is removed quasi-statically. Then, the slope of the $\sigma_x - \varepsilon_x$ diagram during the unloading is measured as the E_x , and then the loading continues. Repeating this procedure up to the rupture and gathering the data will result in the $E_x - \varepsilon$ diagram which for the case study of the paper is as shown in Fig. 1. In Fig. 1, the E_x is normalized by its initial (undamaged) value.

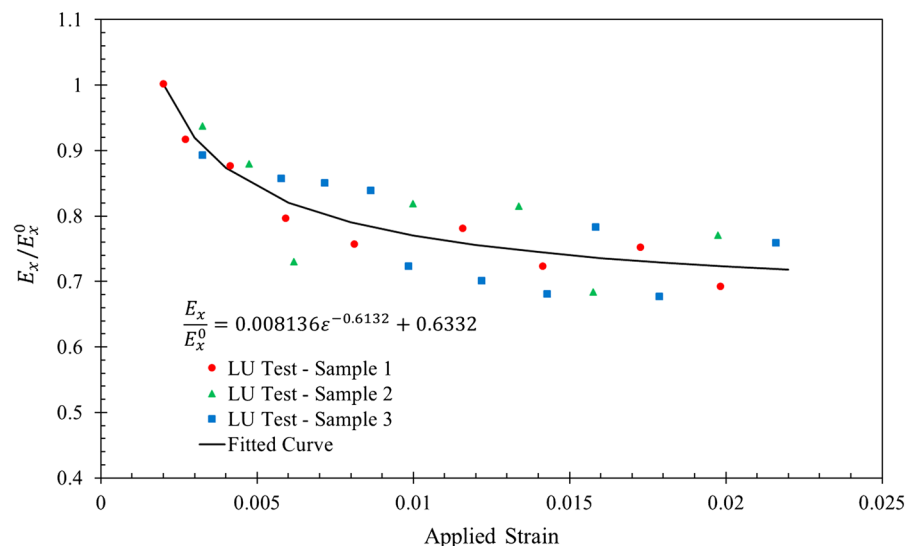
To experimentally study the progress of damage, or in other words, to determine the $1/s - \varepsilon$ diagram, a light-transmission test is applied. According to Fig. 2, the test setup includes a universal servo-electrical testing machine (Santam STM-150), an embedded load cell, an external light source, a GE $[0/90_4]_s$ sample, a digital camera, and the clamps.

The lower grip of the machine is fixed, and the upper one is mobile. The sample is mounted on the grips of the machine. The testing machine loads the sample quasi-statically at a 0.5 mm/min rate, and the embedded load cell reads the loading data. As the load increases, matrix cracks form in the sample. The camera is attached to the fixture using clamps. The light is transmitted through the sample from the light source, and the camera records the shadows of matrix cracks corresponding to each load level (damage progression), as shown in Fig. 3. As such, the number of matrix cracks

Table 1 Mechanical properties of the unidirectional GE ply

E_1 (GPa)	E_2 (GPa)	E_3 (GPa)	ν_{12}	ν_{13}	ν_{23}	G_{12} (GPa)	G_{13} (GPa)	G_{23} (GPa)
34	3.2	3.2	0.27	0.27	0.4	2.3	2.3	1.2

Fig. 1 $E_x/E_x^0 - \varepsilon$ diagram for the GE $[0/90_4]_s$ laminate



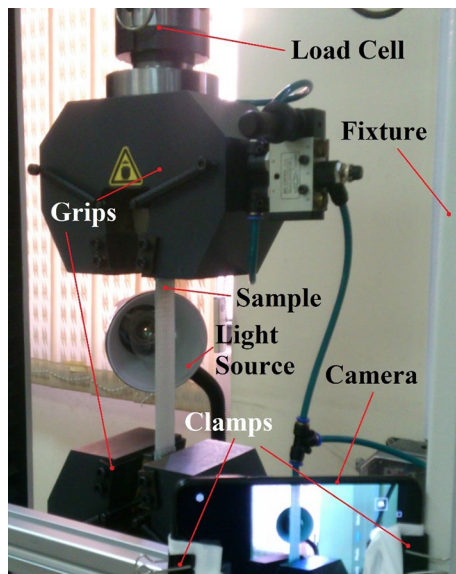


Fig. 2 Light-transmission test setup

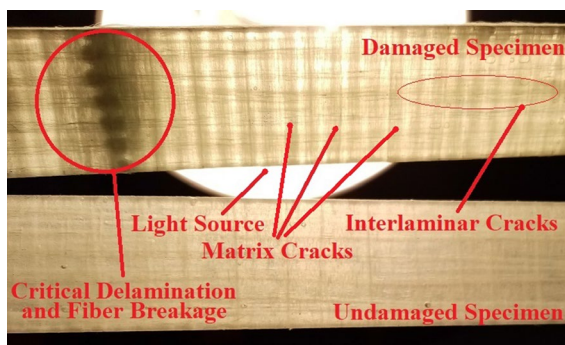


Fig. 3 A sample of recorded video and damage entities

can be obtained in each load level, and as depicted in Fig. 4, the $1/s - \epsilon$ diagram can be specified.

3 FE micromechanical model

3.1 Modeling procedure

The case-study sample is a symmetrical and balanced GE $[0/90_4]_s$ laminate under tensile load in the x -direction (Fig. 5). In this case, no tensile-shear or tensile-bending coupling occurs in the laminate, and a periodic array of the matrix cracks perfectly form

in 90° layers. As such, the equivalent material of the entire laminate with/without damage is an orthotropic material. Therefore, matrix cracks are considered to be equally spaced, similar, straight, and fully grown through the thickness and width of the 90° layers along the fibers. With these assumptions, the composite laminate has a periodic geometry. So, it can be divided into a large number of completely identical unit cells according to Fig. 6. Thus, this cell can be considered a representative volume element (RVE). With this choice, the length of the RVE in a simulation is equal to the distance between two adjacent cracks (s). Therefore, to determine the elasticity modulus in terms of crack density ($1/s$), several simulations with different RVE lengths should be performed. The width of the RVE is set to be 1mm which is high enough to minimize the edge effects.

FE simulations are done in Abaqus commercial software. Each layer is created as a separate three-dimensional deformable part, and then a tie constraint is applied between the adjacent layers. The mechanical properties of GE are entered into the software from Table 1, and 8-node linear brick elements (C3D8) are selected as element type. RVE is a repeating unit cell, so periodic boundary conditions (PBC) should be applied in the simulation. These boundary conditions are as follows:

$$u_i(x_\alpha + \Delta x_\alpha) = u_i(x_\alpha) + \Delta x_\beta \left\langle \frac{\partial u_i}{\partial x_\beta} \right\rangle \quad (1a)$$

$$\epsilon_{ij}(x_\alpha + \Delta x_\alpha) = \epsilon_{ij}(x_\alpha) \quad (1b)$$

$$\sigma_{ij}(x_\alpha + \Delta x_\alpha) = \sigma_{ij}(x_\alpha) \quad (1c)$$

where u_i , ϵ_{ij} , σ_{ij} , Δx_β and $\left\langle \frac{\partial u_i}{\partial x_\beta} \right\rangle$ (with indices $\alpha, \beta, i = 1, 2, 3$) represent the displacement, strain, stress, vector of periodicity, and volume-averaged displacement gradients, respectively. This boundary condition states that the fields of stress, strain, and displacement on the boundaries of RVE should be periodic. Equations (1a) to (1c) are different forms of the same fact, so one needs to apply one of them in the simulation.

PBCs in the form of Eq. (1a) are actually displacement constraints between respective nodes on the respective faces (Fig. 7a). The equations in

Fig. 4 $1/s - \varepsilon$ diagram for the GE $[0/90_4]_s$ laminate

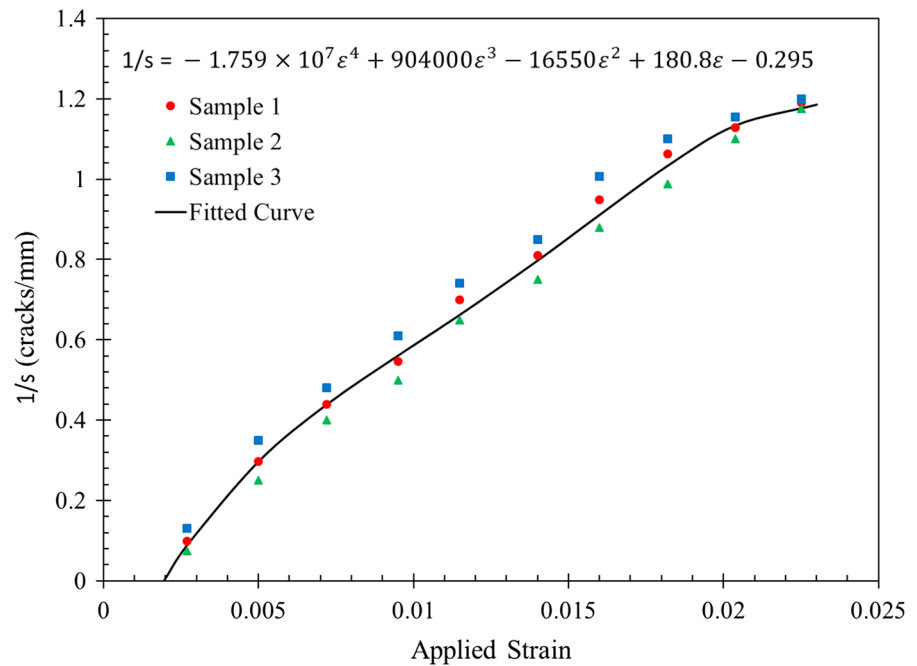


Fig. 5 Schematic of matrix cracks in the GE $[0/90_4]_s$ laminate under tension

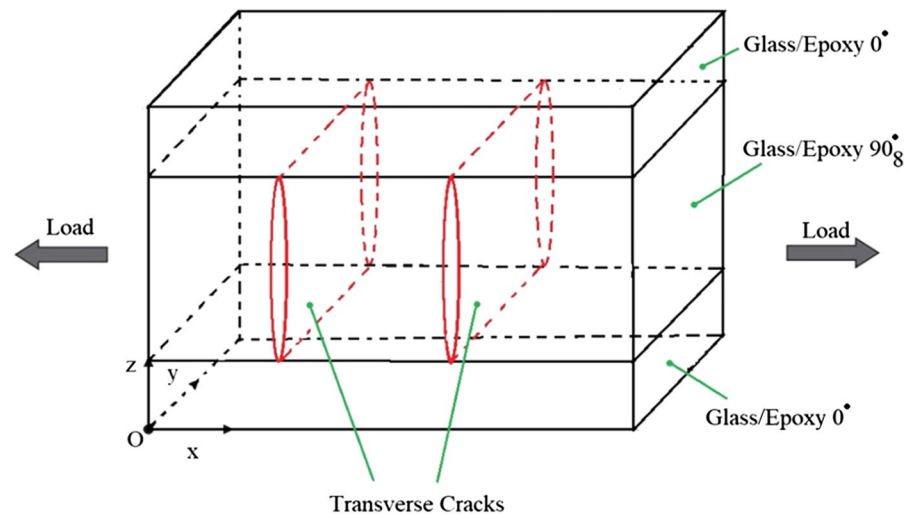


Fig. 7a are derived from Eq. (1a) considering only tension load in the x - or y -directions. As shown in Fig. 7b, PBCs should be applied on the left, right, top, and bottom faces of the RVE. This means that displacement constraints of Fig. 7a, must be applied to all the nodes on these faces. This can be done by using a Python code that labels the nodes and creates the displacement constraints. The boundary condition on the back face of the RVE is traction free. Also, Because of the symmetry,

only half of the RVE is simulated in the Abaqus, and the xy -symmetry boundary condition is held on the front face. The displacements of nodes O, A, B, and D are restricted such that no rigid body displacement/rotation occurs. Since E_x , ν_{xy} , and E_y are going to be obtained, the load is a displacement tension either in the x -direction on the right face for the E_x and ν_{xy} , or in the y -direction on the upper face for the E_y .

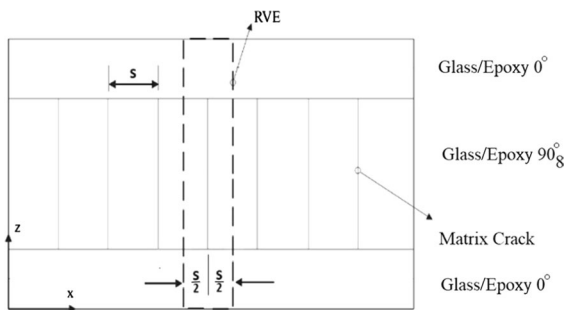


Fig. 6 xz cross-section of the GE $[0/90_4]_s$ laminate RVE

3.2 Results and discussion

Simulations have shown that matrix cracks of the GE $[0/90_4]_s$ laminate will not affect E_y and E_z . So, only E_x changes are investigated in the section. Since calculated E_x and E_x^0 (undamaged E_x) are different from the experimental values, it is convenient to investigate the E_x/E_x^0 in terms of $1/s$ to eliminate the effect of this error. According to Fig. 8, the micromechanical model has a very good accuracy and predicts the normalized elasticity modulus with a maximum error of 6.3%. The reason for the high accuracy is the dominant damage mechanism, i.e., transverse cracks, and the shielding effect of these cracks being considered in the FE model. The micromechanical model always estimates the normalized elasticity modulus more than the experimental data, and as higher is the crack density, so higher is the prediction error; This error comes from unconsidered damage mechanisms, the number and effect of which is greater in higher loadings. However, such high loadings are mostly avoided in real structural applications.

Now that the FE micromechanical model has been validated, we can draw the curve of the elasticity modulus in terms of applied strain, as shown in

Fig. 9. The micromechanical model estimates the value of the pristine elasticity modulus of the GE $[0/90_4]_s$ laminate using the mechanical properties of its constituent unidirectional layers with an error of 3.4%. In the beginning, the cracks are very far apart, and each opens freely due to loading. For this reason, at the beginning of loading and with the appearance of transverse cracks, the elasticity modulus drops sharply. But as the number of cracks increases at higher loads, these cracks get closer to each other, and perturbation of the stress field around a crack affects the opening displacement of the adjacent cracks. This phenomenon (CSE), not only reduces the formation rate of new cracks but also slows down the rate of elasticity modulus reduction. As mentioned earlier, due to the increase in the number and effect of other damage mechanisms that are not included in the model, the error of the model increases for larger loads. So, for the strain close to rupture, i.e., $\epsilon_x = 0.022$, the error reaches 9.6%.

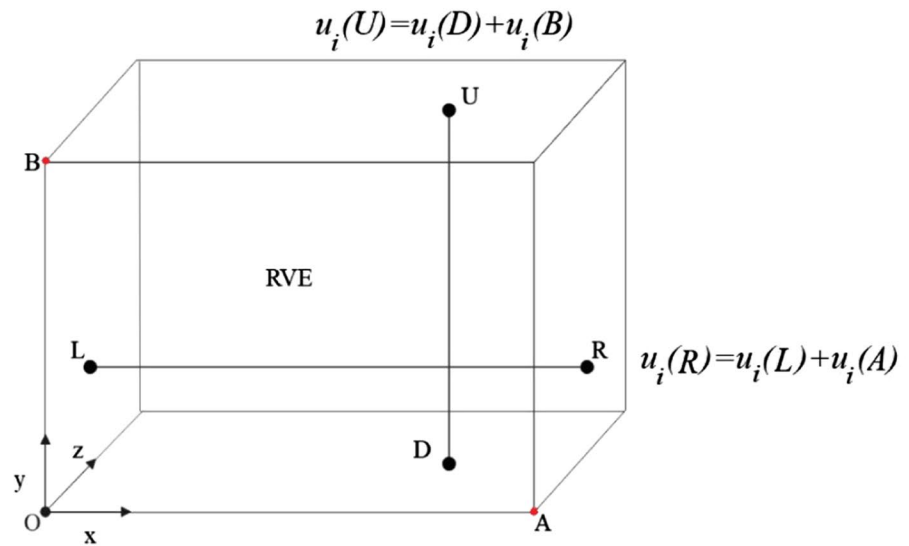
4 CDM model

4.1 Modeling procedure

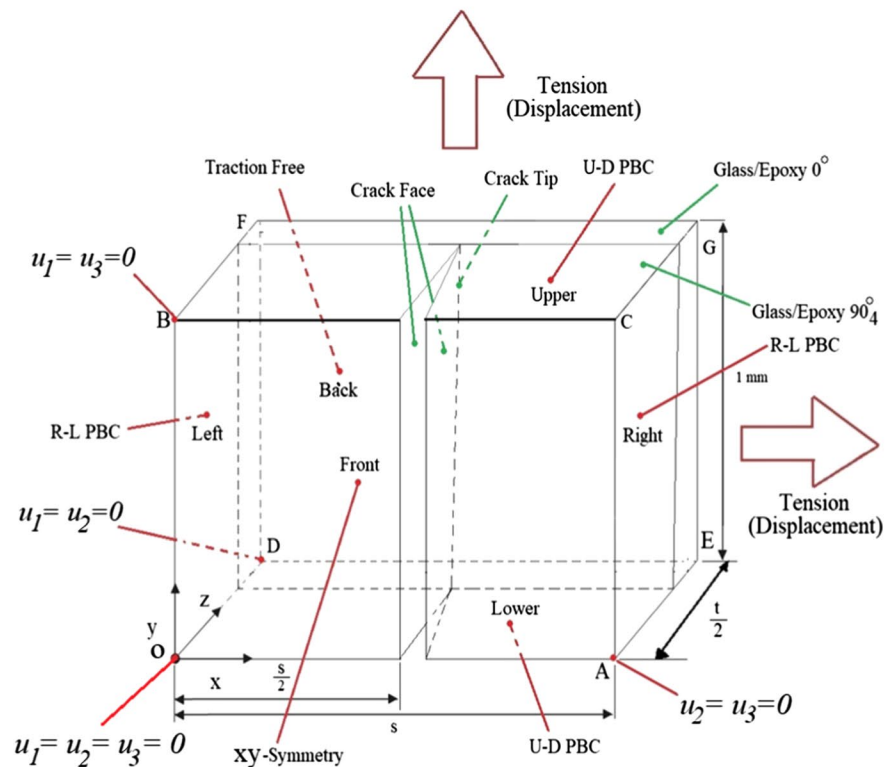
In this section, the CDM modeling of the GE $[0/90_4]_s$ laminate with matrix cracks is explained. The goal here is to determine the equivalent mechanical properties of the entire laminate in terms of crack density $1/s$. For the case-study laminate, matrix cracks will not affect the out-of-plane mechanical properties, so only in-plane properties are investigated. As mentioned in Sect. 3, it is assumed that the matrix cracks are equally spaced, similar, straight, and fully grown through the thickness and width of the 90° layers. Taking inspiration from the models of Talreja and Singh, the new Helmholtz free energy function ψ for a $[0/\pm\theta]_s$ composite laminate contained matrix cracks in $\pm\theta$ layers under isothermal conditions is proposed as follows:

$$\begin{aligned} \psi = & \{c_1\epsilon_1^2 + c_2\epsilon_2^2 + c_3\epsilon_3^2 + c_4\epsilon_1\epsilon_2\} + \{c_5D_1 + c_6D_2\}\epsilon_1^2 + \{c_7D_1 + c_8D_2\}\epsilon_2^2 \\ & + \{c_9D_1 + c_{10}D_2\}\epsilon_3^2 + \{c_{11}D_1 + c_{12}D_2\}\epsilon_1\epsilon_2 + \{c_{13}\epsilon_1\epsilon_3D_3 + c_{14}\epsilon_2\epsilon_3D_3\} \\ & + \{c_{15}D_1^2 + c_{16}D_2^2\}\epsilon_1^2 + \{c_{17}D_1^2 + c_{18}D_2^2\}\epsilon_2^2 + \{c_{19}D_1^2 + c_{20}D_2^2\}\epsilon_3^2 \\ & + \{c_{21}D_1^2 + c_{22}D_2^2\}\epsilon_1\epsilon_2 + \{c_{23}\epsilon_1\epsilon_3D_3^2 + c_{24}\epsilon_2\epsilon_3D_3^2\} + \{c_{25}D_1^3 + c_{26}D_2^3\}\epsilon_1^2 \\ & + \{c_{27}D_1^3 + c_{28}D_2^3\}\epsilon_2^2 + \{c_{29}D_1^3 + c_{30}D_2^3\}\epsilon_3^2 + \{c_{31}D_1^3 + c_{32}D_2^3\}\epsilon_1\epsilon_2 \\ & + \{c_{33}\epsilon_1\epsilon_3D_3^3 + c_{34}\epsilon_2\epsilon_3D_3^3\} \end{aligned} \quad (2)$$

Fig. 7 Boundary conditions of the GE $[0/90_4]_s$ laminate RVE



(a) Periodic boundary conditions of displacement



(b) Applying boundary conditions

where $c_i, i = 1, 2, \dots, 34$ are constants, and $\varepsilon_j, D_j, j = 1, 2, 3$ are strain and damage parameters, respectively, such that

$\varepsilon_1 = \varepsilon_x, \varepsilon_2 = \varepsilon_y, \varepsilon_3 = \varepsilon_{xy}, D_1 = D_{11}, D_2 = D_{22},$ and $D_3 = D_{12}$. Also, matrix crack damage parameters are defined as [12]:

Fig. 8 $E_x/E_x^0 - 1/s$ diagram of the GE $[0/90_4]_s$ laminate

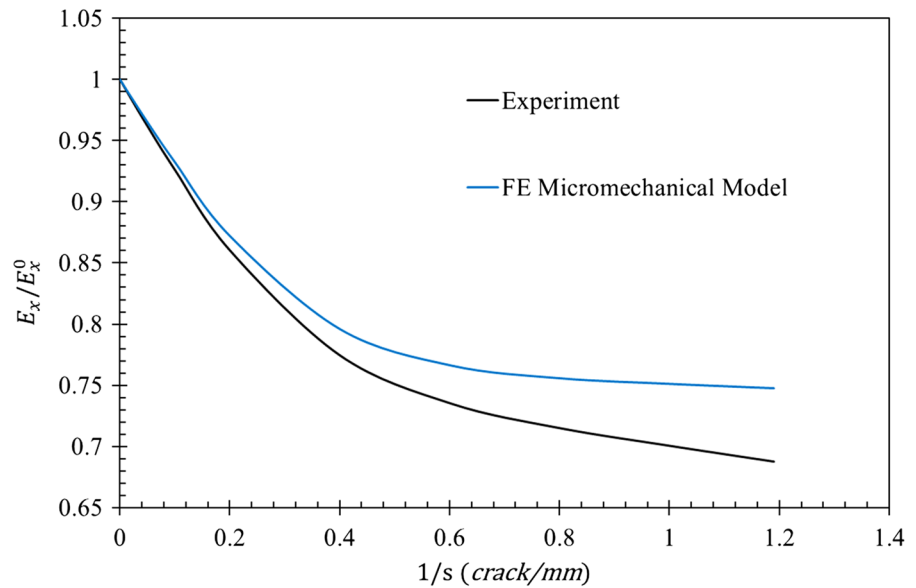
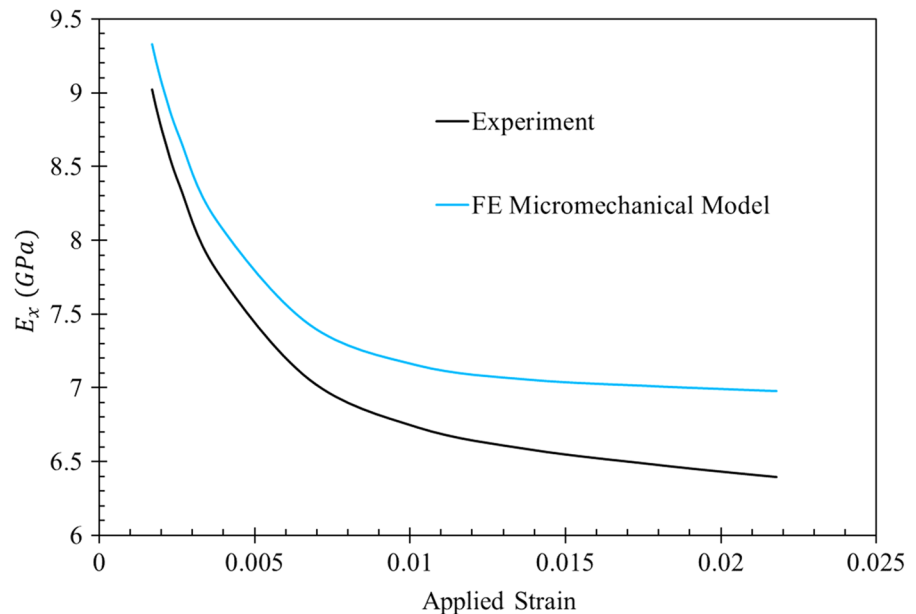


Fig. 9 $E_x - \varepsilon$ diagram of the GE $[0/90_4]_s$ laminate



$$D_{ij} = \frac{\kappa t_c^2}{s t \sin \theta} n_i n_j \quad (3)$$

where $n_i = (\sin \theta, \cos \theta, 0)$ are components of the unit vector normal to a matrix crack plane. Also, κ , t_c , and t are constraint parameter, the thickness of the cracked layers, and laminate thickness, respectively. The constraint parameter represents the constraint effects of surrounding layers on the COD, and is a constant. According to Eq. (3), crack density, $1/s$, can

be used to represent the damage state. From CDM theory, the stiffness matrix can be obtained from Helmholtz free energy:

$$C_{ij} = \frac{\partial^2 \psi}{\partial \varepsilon_i \partial \varepsilon_j} \bigg|_{D_1, D_2, D_3} \quad (4)$$

In Eq. (4), the vertical line symbol and its following parameters, $D_i, i = 1, 2, 3$, imply partial derivative of the Helmholtz energy with respect to the strains, while D_i

are considered constants. The relations between equivalent mechanical properties and stiffness matrix components for an orthotropic laminate are as follows:

$$E_x = \frac{C_{11}C_{22} - C_{12}^2}{C_{22}} \quad (5a)$$

$$E_y = \frac{C_{11}C_{22} - C_{12}^2}{C_{11}} \quad (5b)$$

$$\nu_{xy} = \frac{C_{12}}{C_{22}} \quad (5c)$$

$$G_{xy} = C_{33} \quad (5d)$$

where E , ν , and G are modulus of elasticity, Poisson's ratio, and shear modulus, respectively. As mentioned, for the case-study configuration, laminate is orthotropic with/without damage. As such, using Eqs. (2) to (5), and $\theta = 90^\circ$ the equivalent mechanical properties are derived as follows:

$$E_x = \left(\frac{E_x^0}{1 - \nu_{xy}^0 \nu_{yx}^0} + 2 \frac{\kappa t_c^2}{st} a_1 + 2 \frac{\kappa^2 t_c^4}{s^2 t^2} b_1 + 2 \frac{\kappa^3 t_c^6}{s^3 t^3} f_1 \right) - \left(\frac{\nu_{xy}^0 E_y^0}{1 - \nu_{xy}^0 \nu_{yx}^0} + \frac{\kappa t_c^2}{st} a_4 + \frac{\kappa^2 t_c^4}{s^2 t^2} b_4 + \frac{\kappa^3 t_c^6}{s^3 t^3} f_4 \right)^2 / \left(\frac{E_y^0}{1 - \nu_{xy}^0 \nu_{yx}^0} + 2 \frac{\kappa t_c^2}{st} a_2 + 2 \frac{\kappa^2 t_c^4}{s^2 t^2} b_2 + 2 \frac{\kappa^3 t_c^6}{s^3 t^3} f_2 \right) \quad (6)$$

$$E_y = \left(\frac{E_y^0}{1 - \nu_{xy}^0 \nu_{yx}^0} + 2 \frac{\kappa t_c^2}{st} a_2 + 2 \frac{\kappa^2 t_c^4}{s^2 t^2} b_2 + 2 \frac{\kappa^3 t_c^6}{s^3 t^3} f_2 \right) - \left(\frac{\nu_{xy}^0 E_x^0}{1 - \nu_{xy}^0 \nu_{yx}^0} + \frac{\kappa t_c^2}{st} a_4 + \frac{\kappa^2 t_c^4}{s^2 t^2} b_4 + \frac{\kappa^3 t_c^6}{s^3 t^3} f_4 \right)^2 / \left(\frac{E_x^0}{1 - \nu_{xy}^0 \nu_{yx}^0} + 2 \frac{\kappa t_c^2}{st} a_1 + 2 \frac{\kappa^2 t_c^4}{s^2 t^2} b_1 + 2 \frac{\kappa^3 t_c^6}{s^3 t^3} f_1 \right) \quad (7)$$

$$\nu_{xy} = \left(\frac{\nu_{xy}^0 E_y^0}{1 - \nu_{xy}^0 \nu_{yx}^0} + \frac{\kappa t_c^2}{st} a_4 + \frac{\kappa^2 t_c^4}{s^2 t^2} b_4 + \frac{\kappa^3 t_c^6}{s^3 t^3} f_4 \right) / \left(\frac{E_y^0}{1 - \nu_{xy}^0 \nu_{yx}^0} + 2 \frac{\kappa t_c^2}{st} a_2 + 2 \frac{\kappa^2 t_c^4}{s^2 t^2} b_2 + 2 \frac{\kappa^3 t_c^6}{s^3 t^3} f_2 \right) \quad (8)$$

$$G_{xy} = G_{xy}^0 + 2 \frac{\kappa t_c^2}{st} a_3 + 2 \frac{\kappa^2 t_c^4}{s^2 t^2} b_3 + 2 \frac{\kappa^3 t_c^6}{s^3 t^3} f_3 \quad (9)$$

where superscript 0 denotes the pristine state of the parameters. Also $a_i, b_i, f_i, i = 1, 2, 3, 4$ are damage constants, which are obtained using FE micromechanical models described in Sect. 3. To obtain the constants, one needs to do three FE micromechanical simulations with arbitrary crack densities $1/s_1, 1/s_2, 1/s_3$, and obtain the respective set of parameters $(E_x, E_y, \nu_{xy}, G_{xy})$ from the simulations, each set called a reference state. Then substituting three reference states in Eqs. (6) to (9), damage constants can be determined. Also, the pristine state of the mechanical properties required for the determination of Eqs. (6) to (9) can be determined using the FE models of Sect. 3 without crack.

Damage progression, i.e., crack density evolution in terms of applied strain, was experimentally obtained in Sect. 2. However, it is practical for analytical goals to characterize it using theoretical tools. Singh and Talreja [11] have proposed an energy-based model to characterize the evolution of matrix cracks in multidirectional composite laminates. They have introduced a new parameter W_C called crack multiplication toughness, which is a laminate, and material constant. W_C is interpreted as the total energy dissipated in the completion of the process of crack multiplication from N (with crack spacing s) to $2N$ (with crack spacing $s/2$) cracks. They have suggested the following criterion for crack multiplication:

$$\frac{W}{W_C} \geq 1 \quad (10)$$

where W is the work performed in closure of the new crack planes, which can be calculated as follows:

$$W = \frac{(\sigma_{20}^\theta)^2 t_c}{E_2} \left[2\tilde{u}_n^\theta \left(\frac{s}{2} \right) - \tilde{u}_n^\theta(s) \right] \quad (11)$$

where $\sigma_{20}^\theta, E_2, \tilde{u}_n^\theta$ are the undamaged-state stress normal to the crack plane at the onset of multiplication, transverse modulus of elasticity of the cracked layers,

and COD averaged over the crack surfaces. Using the Fig. 4 and Eqs. (10) and (11), W_C is obtained analytically for the GE $[0/90_4]_s$ laminate. This energy-based model is a part of the SDM methodology presented by Singh and Talreja, which does not get changed from one SDM model to another. So, to avoid duplication, the reader may refer to [11] for the detailed procedure to obtain the W_C .

4.2 Results and discussion

Since Eqs. (6) to (8) are coupled and Eq. (9) and shear loading is not studied in this paper, it is only needed to investigate E_x . Figures 10 and 11 are presented to compare the proposed model with Talreja and Singh's models, as well as verify it with experimental results. As seen in Figs. 10 and 11, Talreja's model in the low crack density/applied strain has good accuracy. But at higher densities/strains, the modulus obtained from Talreja's model is much lower than the

Fig. 10 $E_x/E_x^0 - 1/s$ diagram of the GE $[0/90_4]_s$ laminate

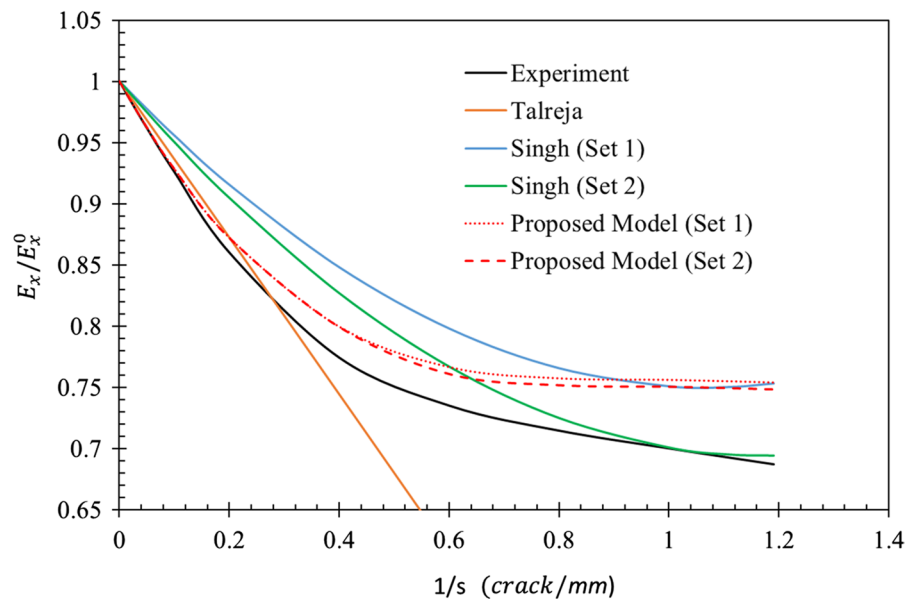
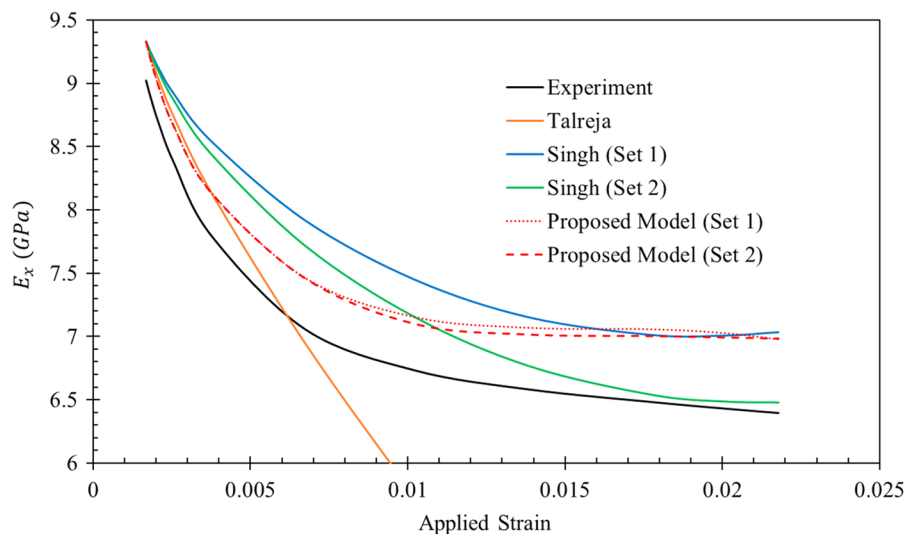


Fig. 11 $E_x - \epsilon$ diagram of the GE $[0/90_4]_s$ laminate



experimental values, so the model error at $1/s = 1.2$ reaches 66.2%. It is because, Talreja's model [7] does not account for CSE, which causes the nonlinear variation of elasticity modulus in terms of crack density/applied strain. To solve this problem, one can use the nonlinear model proposed by Singh [12]. However, Singh claimed that his model is sensitive to the choice of reference states, and there is a possibility of error. Therefore, to check out this claim, the constants are determined from two reference state sets of $1/s(\text{set1}) = \{0.01, 1.11\}$ and $1/s(\text{set2}) = \{0.02, 0.55\}$. As can be seen in Figs. 10 and 11, Singh's model predicts the nonlinear behavior of the elasticity modulus in terms of crack density/applied strain. But as Singh claimed, this model is sensitive to the choice of the reference states, and curves obtained from sets 1 and 2 provide different predictions. Also, the model may result in negative damage prediction error, which can be seen for both sets at the end of both curves. Singh suggested that the reference states should be chosen carefully. But it seems that solving this problem needs more investigation.

The errors of Singh's model arise from the parabolic nature of the model, and the lack of needed reference states (only two) to identify the damage constants. To clarify, Fig. 12 is presented. Singh's experiments [12] showed that for the general symmetric and balanced laminate configuration where transverse cracks form only in one damage mode, i.e., only in layers of $\pm\theta$, the $E_x/E_x^0 - 1/s$ experimental diagram contains three distinct regions; The first one is where the crack formation and multiplication initiate, and cracks are far from each other, so there is no CSE, and E_x/E_x^0 decreases sharply with

an increase of $1/s$. At the beginning of Region 2, there are enough cracks in the sample to generate a weak shielding effect. However, upon the formation of more cracks, the effect gets stronger. As such, the E_x/E_x^0 decreasing rate drops gradually, and Region 2 can be called the transient region. Finally, in Region 3, crack interaction is at full effect, and E_x/E_x^0 is almost constant with the variation of crack density. So, it seems Singh's model with a parabolic nature cannot identify a phenomenon with two turning points.

The new model proposed in this paper contains third-order damage parameters and demands three reference states to obtain its constants. To check out the sensitivity of the proposed model, two reference state sets of $1/s(\text{set1}) = \{0.01, 0.5, 1.11\}$ and $1/s(\text{set2}) = \{0.02, 0.45, 1.19\}$ are used. So as shown in Figs. 10 and 11, it can predict the reduction of stiffness with reasonable accuracy for the full range of the crack density. Its error in predicting undamaged modulus E_x^0 is 3.4%, and its maximum error in predicting E_x at the applied strain $\epsilon = 0.022$ is 9.6%. Also, it is possible to establish reliable criteria for selecting reference states for the proposed model. For Talreja and Singh's models, not only selection criteria are not available but also cannot be established. If the following criteria are met, the prediction of the model is almost coincident with the FE micromechanical results as well as it is almost not sensitive to the reference state selection:

1. A reference state in Region 1 (for the case study: $1/s < 0.2$)
2. A reference state in Region 2 (for the case study: $0.4 < 1/s < 0.6$)
3. A reference state in Region 3 (for the case study: $1/s > 1$)

But, as mentioned earlier, the lack of accuracy of Singh's model is not only from it not being close to experimental data, but also Singh's model will result in negative damage prediction error, which can be seen for both sets at the end of both curves. According to Figs. 10 and 11, the negative damage prediction error of Singh's model is eliminated by the proposed model.

The schematic diagram shown in Fig. 12 is almost valid for any symmetric and balanced thin laminate under tension with matrix cracks of only one damage

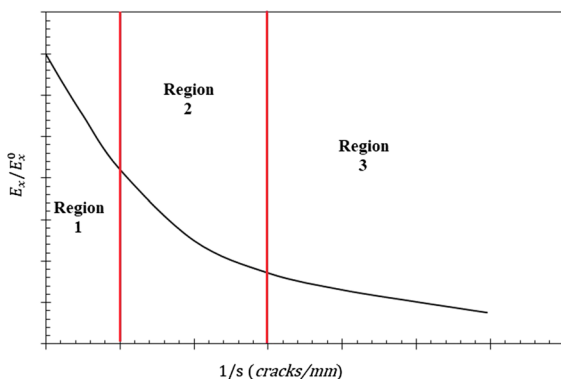


Fig. 12 Schematic $E_x/E_x^0 - 1/s$ diagram

mode. Therefore, the above criteria are valid for all of these laminates.

Since the precision of the proposed model has been validated, according to Fig. 13, it can be used to draw the stress–strain diagram of the GE $[0/90_4]_s$ laminate. According to this figure, the proposed model well predicts the stress–strain curve. Also, showing how much considering the damage of matrix cracks can affect the laminate stress–strain behavior, the diagram of an ideal model, in which no damage exists, is depicted in Fig. 13. The maximum error of both modes is presented in Table 2. The results show that considering the damage of matrix cracks can decrease the error by 37.5% because it well predicts the nonlinear behavior of the stress–strain diagram.

5 Conclusion

In this paper, it was explained why Singh's model leads to errors, and to eliminate the mentioned

errors, a new anisotropic nonlinear damage model was presented and used to predict the stiffness reduction as well as stress–strain behavior of a case-study glass/epoxy $[0/90_4]_s$ composite laminate. The model synergistically applies analytical continuum damage mechanics with an energy-based damage progression model and finite-element micromechanics. The model was validated by experimental tests and compared to Talreja and Singh's models. There was a very good agreement between the experiments and the proposed model. Also, while it was not possible with Talreja and Singh's models, a set of criteria was proposed to select the reference states to achieve the least error and match the result of the analytical model to the result of the FE micromechanical models. In summary, the highlights of the paper are as follows:

1. Abandoned problems that cause errors in Singh's nonlinear model are addressed.

Fig. 13 $\sigma_x - \epsilon_x$ diagram of the GE $[0/90_4]_s$ laminate

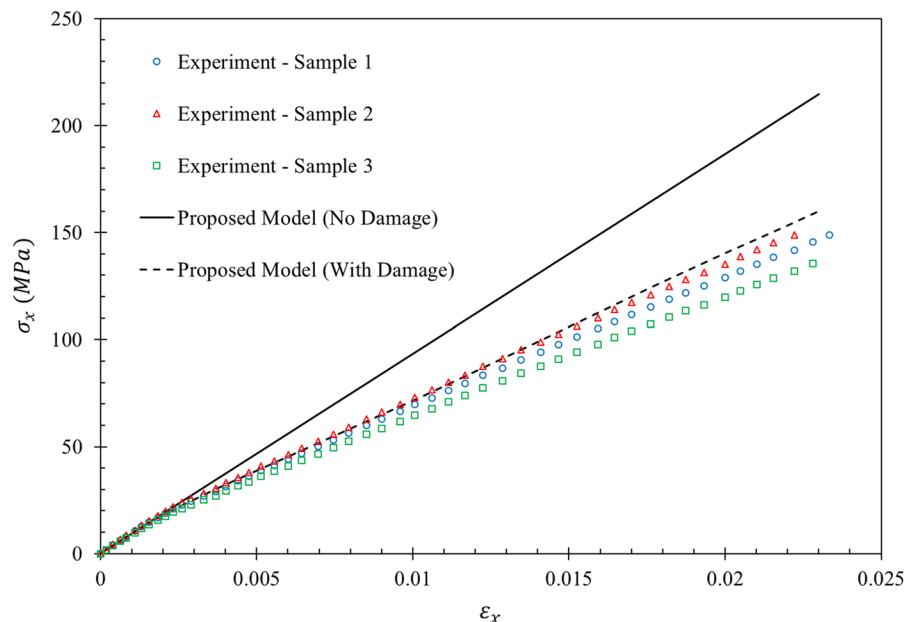


Table 2 Maximum error of the proposed model

Proposed model	Max error at $\epsilon = 0.022$			
	Sample 1	Sample 2	Sample 3	Average
No damage	44.71%	37.82%	55.61%	45.68%
With damage	7.44%(↓ 37.27%)	2.33%(↓ 35.49%)	15.53%(↓ 40.08%)	8.17%(↓ 37.51%)

2. A new anisotropic nonlinear damage model is presented, which does not suffer from errors of Singh's model.
3. It is possible to establish reference state selection criteria for the new model to reach the best outcome, which was not possible with Talreja and Singh's models.
4. The proposed model well predicts the $E_x/E_x^0 - 1/s$, $E_x - \epsilon$, and $\sigma_x - \epsilon_x$ diagrams of the case-study laminate with a maximum error of 6.3%, 9.6%, and 8.17%, respectively.
5. Considering the damage of matrix cracks in the model can increase the accuracy by 37.5%.

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Declarations

Conflict of interest The author declares that he has no known competing financial interests or non-financial interests that could have appeared to influence the work reported in this paper.

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